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a | Hence. $V_R = -8V$; $R_3 = R_F$ Hence, $\frac{R_F}{R_3} = 1$

$$V_o = -\frac{V_R \times R_F}{R} \left(s_3 + \frac{s_2}{2} + \frac{s_1}{4} + \frac{s_0}{8} \right)$$

input case	calculation	output voltage
<u>$s_3 s_2 s_1 s_0$</u>		
0 0 0 0	$8 \times 1 (0 + 0/2 + 0/4 + 0/8)$	0V
0 0 0 1	$8 \times 1 (0 + 0/2 + 0/4 + 1/8)$	1V
0 0 1 0	$8 \times 1 (0 + 0/2 + 1/4 + 0/8)$	2V
0 0 1 1	$8 \times 1 (0 + 0/2 + 1/4 + 1/8)$	3V
0 1 0 0	$8 \times 1 (0 + 1/2 + 0/4 + 0/8)$	4V
0 1 0 1	$8 \times 1 (0 + 1/2 + 0/4 + 1/8)$	5V
0 1 1 0	$8 \times 1 (0 + 1/2 + 1/4 + 0/8)$	6V
0 1 1 1	$8 \times 1 (0 + 1/2 + 1/4 + 1/8)$	7V
1 0 0 0	$8 \times 1 (1 + 0/2 + 0/4 + 0/8)$	8V
1 0 0 1	$8 \times 1 (1 + 0/2 + 0/4 + 0/8)$	9V
1 0 1 0	$8 \times 1 (1 + 0/2 + 1/4 + 0/8)$	10V
1 0 1 1	$8 \times 1 (1 + 0/2 + 1/4 + 1/8)$	11V
1 1 0 0	$8 \times 1 (1 + 1/2 + 0/4 + 0/8)$	12V
1 1 0 1	$8 \times 1 (1 + 1/2 + 0/4 + 1/8)$	13V

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$$1110 \quad 8 \times 1 (1 + 1/2 + 1/4 + 0/8) \quad 14V$$

$$1111 \quad 8 \times (1 + 1/2 + 1/4 + 1/8) \quad 15V$$

b) here, maximum output voltage = 15V
LSB voltage = 1V

$\therefore$ ratio of max. output voltage
and LSB voltage is $\frac{15}{1} = 15$.

c) if $V_0 = 5V$;

$$V_0 = -\frac{V_R \times R_f}{R_3} \left(s_3 + \frac{s_2}{2} + \frac{s_1}{4} + \frac{s_0}{8} \right)$$

$$\Rightarrow 5 = \frac{8 \times R_f}{R_3} (1 + 0/2 + 1/4 + 0/8)$$

$$\Rightarrow 5 = 8 \times 1/2 \times \left(\frac{R_f}{R} \right)$$

$$\Rightarrow 5 = 10 \times \frac{R_f}{R_3}$$

$$\Rightarrow \frac{R_f}{R_3} = 0.5$$

$$\Rightarrow \frac{R_f}{R} =$$

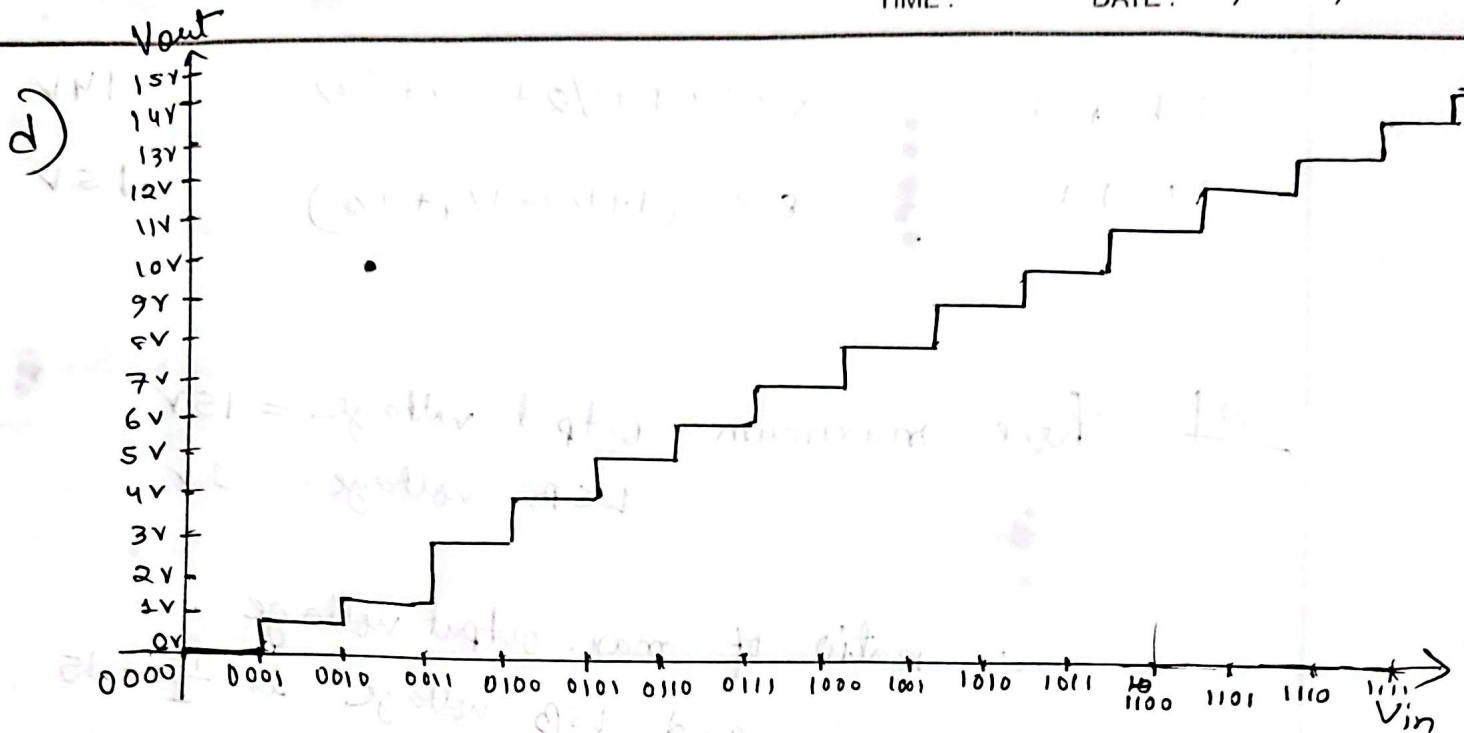
$$\Rightarrow \frac{R_3}{R_f} = 2$$

here, $R_3/R_f = 2$. The ratio is 2.

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e) To improve the resolution of the DAC we can -

- ① increase the number of bits $\rightarrow$ more input combinations will have finer steps (which will give higher resolution).
- ② Use R-2R Ladder instead of binary weighted DAC $\rightarrow$ R-2R Ladder simplifies resistor precision issues which binary weighted DAC lacks of.